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IMAGE FORMING APPARATUS

Abstract

An image forming apparatus including an image carrier, a long print head including a board and a lens array, and a tilt adjuster that adjusts a tilt of the print head, further includes a positioning mechanism that positions the print head relative to the image carrier to maintain a constant distance d between the print head and the image carrier, regardless of the tilt of the print head about a swing axis β caused by the tilt adjuster.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority from Japanese Application JP 2024-037237, the content to which is hereby incorporated by reference into this application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present disclosure relates to an image forming apparatus such as a copier, a multifunction peripheral, a printer, and a facsimile device.

2. Description of the Related Art

[0003] In the related art, there is known an image forming apparatus including an image carrier (for example, a photosensitive drum), a long print head, and a tilt adjuster that adjusts a tilt of a print head about a swing axis along an optical axis direction of a plurality of light-emitting elements.

SUMMARY OF THE INVENTION

[0004] The well-known image forming apparatus including the tilt adjuster of the print head has the following disadvantages. The disadvantages will be explained below with reference to FIGS. **20A** to **20C**.

[0005] FIGS. **20A** to **20C** are plan views (top) and front views (bottom) illustrating one example, another example, and yet another example of a distance d (d_1 , d_2 , d_3) between a photosensitive drum **1** and a print head **3** in a well-known image forming apparatus **100X** including a tilt adjuster of the print head **3**. In FIGS. **20A** to **20C**, a symbol FL indicates a main body frame.

[0006] As illustrated in FIGS. **20A** to **20C**, the long print head **3** includes a plurality of light-emitting elements **31** to **31** arranged in a line along a longitudinal direction L, which emit light to expose the photosensitive drum **1** (an example of an image carrier). A lens array **32** focuses light emitted by the plurality of light-emitting elements **31** to **31** onto the photosensitive drum **1**.

[0007] In such a well-known image forming apparatus **100X**, as illustrated in FIGS. **20B** and **20C**, skewing and the like of a sheet P such as a recording sheet may occur. The tilt adjuster adjusts a tilt of the print head **3** around a swing axis β along an optical axis direction N of the plurality of light emitting elements **31** to **31** in accordance with the skewing and the like of the sheet P or the like. In such an example, the swing axis β is located outside the print head **3** in a rotation axis direction X of the photosensitive drum **1**.

[0008] A focal depth of the print head **3** serving as an exposure device using the plurality of light-emitting elements **31** to **31** and the lens array **32**, is a value (for example, about 0.1 mm) smaller than that of another type of exposure device (for example, an optical scanning device being an exposure device using a light deflector and an f θ lens). Thus, it is necessary to position the print head **3** relative to the photosensitive drum **1** so that a distance between the photosensitive drum **1** and the print head **3** is constant.

[0009] However, in the well-known image forming apparatus **100X**, in adjusting a tilt of the print head **3** around the swing axis β , the print head **3** moves parallel to a linear direction T perpendicular to both the rotation axis α and the swing axis β as viewed from the rotation axis direction X of the photosensitive drum **1**, so that the distance d (d_1 , d_2 , d_3) between the photosensitive drum **1** and the print head **3** varies depending on a position of the print head **3** in the linear direction T ($d_1 < d_2 < d_3$). For example, if the focal depth is 0.1 mm, even when a focus of the lens array **32** on a surface **1a** of the photosensitive drum **1** is adjusted to match the configuration illustrated in FIG. **20A**, the distance d (d_2) between the photosensitive drum **1** and the print head **3** in the configuration illustrated in FIG. **20B**, which is moved horizontally by +1 mm (to the left in an example of FIG. **20A**), is $d_1 + 0.258$ mm. That is, as illustrated in FIGS. **20B** and **20C**, as the print head **3** moves more in the linear direction T, the distance d ($d_1 < d_2 < d_3$) between the photosensitive drum **1** and the print head **3** is greater, and thus, the image formed on the surface **1a** of the photosensitive drum **1** is out of focus. Such a phenomenon is more pronounced as the diameter (curvature) of the photosensitive drum **1** is smaller as the image forming apparatus **100X** is more

compact.

[0010] Therefore, an object of the present disclosure is to provide an image forming apparatus capable of maintaining a constant distance between a print head and an image carrier, regardless of a tilt of the print head around a swing axis caused by a tilt adjuster.

[0011] In order to solve the above problem, an image forming apparatus according to one aspect of the present disclosure is an image forming apparatus including an image carrier, a print head having a long length and including a board including a plurality of light-emitting elements arranged in a line and a lens array that focuses light emitted by the plurality of light-emitting elements onto the image carrier, and a tilt adjuster that adjusts a tilt of the print head about a swing axis along an optical axis direction of the plurality of light-emitting elements, in which the image forming apparatus further includes a positioning mechanism that positions the print head relative to the image carrier to maintain a constant distance between the print head and the image carrier, regardless of the tilt of the print head about the swing axis caused by the tilt adjuster.

[0012] According to the present disclosure, it is possible to maintain a constant distance between a print head and an image carrier, regardless of a tilt of a print head about a swing axis caused by a tilt adjuster.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 is a cross-sectional view illustrating an image forming apparatus according to the present embodiment.

[0014] FIG. 2 is a perspective view illustrating an example of a print head and an example of a tilt adjustment mechanism provided on a main body frame in the image forming apparatus according to the present embodiment, as viewed obliquely from above on a front side.

[0015] FIG. 3 is a perspective view of the print head illustrated in FIG. 2 being supported by a swinger, as viewed obliquely from above on the front side.

[0016] FIG. 4A is a perspective view of the print head as viewed obliquely from above on a back side.

[0017] FIG. 4B is a perspective view of the print head illustrated in FIG. 4A as viewed obliquely from below on the front side.

[0018] FIG. 4C is an exploded perspective view of the print head illustrated in FIG. 4A as viewed obliquely from above on the front side.

[0019] FIG. 4D is an exploded perspective view of the print head illustrated in FIG. 4A as viewed obliquely from below on the back side.

[0020] FIG. 5A is a cross-sectional perspective view of the print head as viewed obliquely from above on the front side.

[0021] FIG. 5B is a cross-sectional view of the print head.

[0022] FIG. 5C is a schematic diagram clearly illustrating the cross-sectional view illustrated in FIG. 5B.

[0023] FIG. 5D is a plan view illustrating a state in which a flexible circuit board extends in a straight line in a light-emitting element board, the flexible circuit board, and a printed wiring board included in the print head.

[0024] FIG. 6A is a perspective view of a swing supporter provided in the swinger, as viewed obliquely from above on the back side.

[0025] FIG. 6B is a perspective view illustrating a state in which the swing supporter provided in the swinger is inserted into a through hole in a frame on the back side, as viewed obliquely from above on the back side.

[0026] FIG. 6C is an exploded perspective view of the swinger, the swing supporter, and the

through hole in a main body frame, as viewed obliquely from below on the back side.

[0027] FIG. 7A is a perspective view of a tilt adjuster provided in the main body frame on the front side, as viewed obliquely from above on the front side.

[0028] FIG. 7B is a perspective view of the tilt adjuster provided in the main body frame on the front side, as viewed obliquely from below on the back side.

[0029] FIG. 7C is an exploded perspective view of the tilt adjuster as viewed obliquely from above on the front side.

[0030] FIG. 7D is an exploded perspective view of the tilt adjuster as viewed obliquely from above on the back side.

[0031] FIG. 8 is a perspective view of a photosensitive unit, the print head, and the swinger as viewed from above on a left side.

[0032] FIG. 9A is a perspective view of a portion of a positioner on one side of the swinger including the print head, as viewed obliquely from above.

[0033] FIG. 9B is an exploded perspective view of the print head and the swinger illustrated in FIG. 9A, as viewed obliquely from above.

[0034] FIG. 9C is an exploded perspective view of the print head and the swinger illustrated in FIG. 9A, as viewed obliquely from below.

[0035] FIG. 9D is a cross-sectional perspective view of the positioner and an area to be positioned on one side illustrated in FIG. 8, as viewed obliquely from above.

[0036] FIG. 9E is an enlarged view of the positioner and the area to be positioned on one side in a $\gamma 1$ portion illustrated in FIG. 8.

[0037] FIG. 10A is a perspective view of a portion of the positioner on the other side in the swinger including the print head, as viewed obliquely from above.

[0038] FIG. 10B is an exploded perspective view of the print head and the swinger illustrated in FIG. 10A, as viewed obliquely from above.

[0039] FIG. 10C is an exploded perspective view of the print head and the swinger illustrated in FIG. 10A, as viewed obliquely from below.

[0040] FIG. 10D is a perspective view of the positioner and the area to be positioned on the other side illustrated in FIG. 8, as viewed obliquely from below.

[0041] FIG. 10E is an enlarged view of the positioner and the area to be positioned on the other side in a $\gamma 2$ portion illustrated in FIG. 8.

[0042] FIG. 11 is a cross-sectional view illustrating the positioner on the one side abutting against the area to be positioned on the one side.

[0043] FIG. 12 is a front view for explaining how a distance between the print head and the photosensitive drum is maintained constant, regardless of a tilt of the print head around the swing axis.

[0044] FIG. 13A is an exploded perspective view of the positioner on the one side and a lens holding member, as viewed obliquely from below.

[0045] FIG. 13B is a perspective view illustrating the positioner on the one side attached to the lens holding member, as viewed obliquely from below.

[0046] FIG. 14A is a perspective view of the positioner on the one side as viewed from above on the back side.

[0047] FIG. 14B is a perspective view of the positioner on the one side as viewed from below on the front side.

[0048] FIG. 15A is an exploded perspective view of the positioner on the other side and the lens holding member, as viewed obliquely from below.

[0049] FIG. 15B is a perspective view illustrating the positioner on the other side attached to the lens holding member, as viewed obliquely from below.

[0050] FIG. 16A is a perspective view of the positioner on the other side as viewed from below on the back side.

[0051] FIG. **16B** is a perspective view of the positioner on the other side as viewed from below on the front side.

[0052] FIG. **17** is a side view illustrating another example of a positioning mechanism.

[0053] FIG. **18A** is a front view of a guide mechanism illustrated in FIG. **17**.

[0054] FIG. **18B** is a rear view of the guide mechanism illustrated in FIG. **17**.

[0055] FIG. **19** is a front view for explaining that a distance between the print head and the photosensitive drum is maintained constant regardless of a tilt of the print head around the swing axis, and the optical axes of the plurality of light-emitting elements in the print head orient toward the center of rotation of the photosensitive drum.

[0056] FIG. **20A** is a plan view and a front view illustrating an example of a distance between a photosensitive drum and a print head in a known image forming apparatus including a tilt adjuster of the print head.

[0057] FIG. **20B** is a plan view and a front view illustrating another example of the distance between the photosensitive drum and the print head in the known image forming apparatus including the tilt adjuster of the print head.

[0058] FIG. **20C** is a plan view and a front view illustrating yet another example of the distance between the photosensitive drum and the print head in the known image forming apparatus including the tilt adjuster of the print head.

DETAILED DESCRIPTION OF THE INVENTION

[0059] Embodiments according to the present disclosure will be described below with reference to the drawings. In the following description, the same parts are denoted by the same reference numerals. The names and functions of the same parts are also the same. Therefore, detailed descriptions of the same parts will not be repeated.

Image Forming Apparatus

[0060] FIG. **1** is a cross-sectional view illustrating an image forming apparatus **100** according to the present embodiment. In the drawings, hereinafter, X is a rotation axis direction of a photosensitive drum **1**, X1 is a front side, X2 is a back side, Y is a left-right direction perpendicular to the rotation axis direction X, Y1 is a right side, Y2 is a left side, and Z is an up-down direction perpendicular to the rotation axis direction X and the left-right direction Y.

[0061] Image data processed in an image forming apparatus main body **101** of the image forming apparatus **100** corresponds to a color image using colors including black (K), cyan (C), magenta (M), and yellow (Y), or a monochrome image using a single color (for example, black). Thus, four of each of the photosensitive drum **1** (an example of an image carrier), a charger **2**, a print head **3**, a developing device **4**, a primary transfer device **5**, and a drum cleaning device **6** are provided to form four types of toner images corresponding to each color, and four image stations Pa, Pb, Pc, and Pd are formed, each corresponding to black, cyan, magenta, and yellow.

[0062] In each of the image stations Pa, Pb, Pc, and Pd, the chargers **2** to **2** uniformly charge the surfaces of the photosensitive drums **1** to **1** driven to rotate in a predetermined rotation direction R, to a predetermined potential. The print heads **3** to **3** expose the surfaces of the photosensitive drums **1** to **1** to form electrostatic latent images on the surfaces **1a** of the photosensitive drums **1** to **1**. The developing device **4** develops the electrostatic latent images on the surface **1a** of the photosensitive drums **1** to **1** to form a toner image on the surface **1a** of the photosensitive drums **1** to **1**. As a result, the toner image of each color is formed on the surface **1a** of each of the photoconductor drums **1** to **1**. The drum cleaning devices **6** to **6** remove and collect residual toner from the surfaces of the photosensitive drums **1** to **1**. The primary transfer devices **5** to **5** sequentially superimpose and transfer the toner image of each color on the surface of each of the photosensitive drums **1** to **1** onto an intermediate transfer belt **23** moved to be revolved by a driving roller **21** and a driven roller **22** in a belt drive device **20**, to form a color toner image on the intermediate transfer belt **23**. The belt cleaning device **7** removes and collects a residual toner from the intermediate transfer belt **23**.

[0063] A transfer nip TN is formed between the intermediate transfer belt **23** and a transfer roller

81 of a secondary transfer device **8**. The transfer roller **81** of the secondary transfer device **8** pinches a sheet P, such as a recording sheet, transported via a sheet transport path **11** in the transfer nip TN and transports the sheet P together with the intermediate transfer belt **23** while transferring the color toner image on the surface of the intermediate transfer belt **23** onto the sheet P. A fixing device **9** sandwiches the sheet P between a fixing member (in such an example, a fixing belt **91**) and a pressure member (in such an example, a pressure roller **92**) and applies heat and pressure to fix the color toner image on the sheet P.

[0064] The sheet P is pulled out from a paper feed cassette **13** by a pickup roller **12**, transported through the sheet transport path **11**, passes through the secondary transfer device **8** and the fixing device **9**, and is discharged to a discharge tray **15** via a discharge rollers **14**. In the sheet transport path **11**, a registration roller **16** and the like are arranged. The registration roller **16** temporarily stops the sheet P, aligns the leading edge of the sheet P, and then starts transporting the sheet P in accordance with a timing for transferring the toner image at the transfer nip TN between the intermediate transfer belt **23** and the transfer roller **81**.

Present Embodiment

[0065] FIG. **2** is a perspective view of an example of the print heads **3** to **3** and an example of a tilt adjustment mechanism **200** provided in a main frame FL (FL**1** to FL**3**) in the image forming apparatus **100** according to the present embodiment, as viewed obliquely from above on the front side X1. FIG. **3** is a perspective view of each of the print head **3** illustrated in FIG. **2** being supported by a swinger **210**, as viewed obliquely from above on the front side X1. FIG. **4A** is a perspective view of the print head **3** as viewed obliquely from above on the back side X2. FIG. **4B** is a perspective view of the print head **3** illustrated in FIG. **4A** as viewed obliquely from below on the front side X1. FIGS. **4C** and **4D** are exploded perspective views of the print head **3** illustrated in FIG. **4A**, as viewed obliquely from above on the front side X1 and obliquely from below on the back side X2, respectively.

[0066] FIG. **5A** is a perspective view of the print head **3**, as viewed obliquely from above on the front side X1. FIG. **5B** is a cross-sectional view of the print head **3**. FIG. **5C** is a schematic diagram clearly illustrating the cross-sectional view illustrated in FIG. **5B**. FIG. **5D** is a plan view illustrating a state in which a flexible circuit board **312a** extends in a straight line in a light-emitting element board **311**, the flexible circuit board **312a**, and a printed wiring board **312b** included in the print head **3**.

[0067] The print heads **3** to **3**, the swingers **210** to **210**, and the like have the same configuration respectively, thus, the print heads **3** to **3** are illustrated in a single drawing in FIGS. **3** to **5D**, and in the following description, each of the print heads **3** to **3** and the swingers **210** to **210** will simply be referred to as the print head **3** and the swinger **210**.

[0068] In the following description, the plurality of light-emitting elements **31** to **31** in the print heads **3** to **3** are each an organic light-emitting diode (OLED), hereinafter simply referred to as OLED. The light-emitting elements **31** to **31** are not limited to OLEDs, but may be light-emitting elements such as inorganic light-emitting diodes (LEDs) or nano-light-emitting diodes (nanoscale light-emitting diodes), which are light-emitting diodes (LEDs) having a size smaller than a micrometer.

[0069] The image forming apparatus **100** according to the present embodiment includes the photosensitive drum **1**, the print head **3** having a long length, and the tilt adjustment mechanism **200** (an example of a tilt adjustment mechanism) (see FIG. **2**). The print head **3** includes the light-emitting element board **311** (an example of a board) and the lens array **32**. The light-emitting element board **311** includes the plurality of light emitting elements **31** to **31** (see FIGS. **5B** to **5D**). The light emitting elements **31** to **31** emit light for exposing the surface **1a** of the photosensitive drum **1**. The light emitting elements **31** to **31** are arranged next to each other and in a line in the longitudinal direction L of the print head **3**. In such an example, the light-emitting element board **311** is a film-like board (flexible OLED panel) mounted with OLED elements (**31** to **31**). This

makes it possible to make the print head **3** smaller and less expensive.

[0070] The lens array **32** extends in the longitudinal direction L and focuses light emitted by the light-emitting elements **31** to **31** onto the surface **1a** of the photosensitive drum **1**. The lens array **32** is arranged to face the light emitting elements **31** to **31**. The tilt adjustment mechanism **200** adjusts a tilt of the print head **3** around the swing axis β (see FIG. **3**) in which the tilt is along the optical axis direction N of the plurality of light emitting elements **31** to **31**. More specifically, the tilt adjustment mechanism **200** adjusts an exposure position of the light emitting elements **31** to **31** relative to the surface **1a** of the photosensitive drum **1** by swinging one side L1 around the swing axis β on another side L2 in the longitudinal direction L of the print head **3**. In such an example, the swing axis β is located outside the print head **3** in the rotation axis direction X.

Print Head

[0071] As illustrated in FIGS. **5A** to **5D**, the print head **3** further includes a driving member **312**, a lens holding member **313**, a fixing member **314** (base member), and a main body member **315**. The driving member **312** drives the light emitting elements **31** to **31**. The lens holding member **313** holds the lens array **32**. The fixing member **314** fixes the lens holding member **313** while the light-emitting element board **311** is fixed. The main body member **315** holds the driving member **312**, the lens holding member **313**, and the fixing member **314**.

[0072] In such an example, the driving member **312** includes the flexible circuit board **312a**, the printed wiring board (PWB) **312b**, a driving circuit element **312c**, and a protective member **312d**. One end of the flexible circuit board **312a** is connected to one end of the light-emitting element board **311**, and the other end of the flexible circuit board **312a** is connected to one end of the printed wiring board **312b**. The flexible circuit board **312a** is a flexible circuit board (Chip on Film (COF)) in which a drive circuit element **312c** (driver IC) that drives the OLED elements (**31** to **31**) on the light-emitting element board **311** is mounted on a film. The printed wiring board **312b** connected to the flexible circuit board **312a** includes an input terminal **312b1** (see FIG. **5C**), and is connected to a connector CN (see FIG. **5C**) via the input terminal **312b1**. The protective member **312d** protects an exposed portion of the light-emitting element board **311** and the flexible circuit board **312a**. The connector CN is connected to an image processing device (not illustrated) provided in the image forming apparatus main body **101**. FIG. **5C** illustrates a state in which the light-emitting elements **31** to **31** are provided to protrude from the outer surface of the light-emitting element board **311** (a panel being an example of a board), which is for the purpose of making the light-emitting elements **31** to **31** easier to be understood, but in reality, as illustrated in FIG. **5B**, the light-emitting elements **31** to **31** are provided inside the panel (**311**) not to protrude from the outer surface of the panel (**311**) (for example, by vapor deposition).

[0073] The lens holding member **313** is a frame-shaped member surrounding outer peripheral surfaces **32a** to **32a** of the lens array **32** along the optical axis direction N (see FIGS. **4C** and **4D**). Here, the optical axis direction N is a direction (thickness direction) perpendicular to both the longitudinal direction L and a width direction M of the print head **3**. The lens array **32** is inserted into an inner surface **313a** of the lens holding member **313** that is a frame-shaped member. The lens array **32** includes at least a portion of the outer peripheral surfaces **32a** to **32a** (in such an example, side surfaces **32a1** and **32a1** along the longitudinal direction L) held by the lens holding member **313**, and the outer peripheral surfaces **32a** to **32a** are bonded at least at some locations (in such an example, a plurality of locations) with an adhesive (for example, an ultraviolet-curing adhesive). Further, a periphery of a bond area between the lens array **32** and the lens holding member **313** is sealed with a caulking agent (caulking resin). This effectively prevents a foreign matter such as dust from entering the lens holding member **313** through the bond area between the lens array **32** and the lens holding member **313**.

[0074] The fixing member **314** is formed in a rectangular parallelepiped shape. A rear surface **313c** of the light-emitting element board **311** (a panel including the light emitting elements **31** to **31**) opposite the light emitting elements **31** to **31** is bonded to a surface **314a** of the fixing member **314**

on a side of the photosensitive drum 1 (one side N1) in the optical axis direction N by an adhesive member E (for example, double-sided adhesive tape) (see FIG. 5C). The rear surface 313c of the lens holding member 313 is fixed to a board surface 311a of the light-emitting element board 311 opposite to the fixing member 314 (a panel including the light emitting elements 31 to 31). A periphery of a bond area between the lens holding member 313 and the light-emitting element board 311 (a panel including the light emitting elements 31 to 31) is sealed with a caulking agent (caulking resin). This effectively prevents a foreign matter such as dust from entering the lens holding member 313 through the bond area between the lens holding member 313 and the light-emitting element board 311 (a panel including the light emitting elements 31 to 31).

[0075] The relative positions (positions in the longitudinal direction L, in the width direction M, and in the optical axis direction N) of the plurality of light-emitting elements 31 to 31 and the lens array 32 are positioned and adjusted in advance in the production process using a jig or the like.

[0076] The main body member 315 includes a placement area 315a, a bent area 315b, and a board guide 315c. The placement area 315a, the bent area 315b, and a board holder 315d are integrally formed.

[0077] The placement area 315a includes the placed fixing member 314 of the print head 3. The placement area 315a includes a placement surface 315a1 on which the fixing member 314 is placed so that a rear surface 314b of the fixing member 314 on the side of the swinger 210 comes into contact with the placement surface 315a1. This allows the fixing member 314 to be reliably placed on the placement surface 315a1 of the placement area 315a. A plurality of (two) positioning protrusions or recesses (positioning protrusions 315a2 and 315a2 in such an example) (see FIG. 4C) are provided on the placement surface 315a1 of the placement area 315a. The rear surface 314b of the fixing member 314 includes positioning recesses or protrusions (positioning recesses 314c and 314c in such an example) (see FIG. 4D) corresponding to the plurality of (two) positioning protrusions or recesses (315a2, 315a2). This allows the fixing member 314 to be reliably positioned relative to the main body member 315. The placement area 315a includes a plurality of (two) holders 315a3 and 315a3 (see FIGS. 4C, 4D, and 5B). The placement area 315a and the holders 315a3 and 315a3 are integrally formed. The holders 315a3 and 315a3 detachably hold the fixing member 314 on the placement area 315a. The holders 315a3 and 315a3 restrict movement of the fixing member 314 towards the light emitting elements 31 to 31 in the optical axis direction N (one side N1). The holders 315a3 and 315a3 includes a locker 315a31 (see FIG. 5B) that locks the surface 314a of the fixing member 314.

[0078] The bent area 315b is bent at an acute angle (for example, less than 60 degrees, about 50 degrees in such an example) relative to the placement area 315a. The printed wiring board 312b is fixed to a surface 315b1 (inner surface) of the bent area 315b facing the placement area 315a. The input terminal 312b1 is provided at an entrance end of a space SP between the placement area 315a and the bent area 315b of the printed wiring board 312b, and the connector CN is connected toward the space SP.

[0079] The board guide 315c includes a curved area 315c1 that folds back the flexible circuit board 312a connected to the light-emitting element board 311 toward the space SP. The curved area 315c1 is curved to fold back at an angle of 180 degrees or more between the light-emitting element board 311 and the flexible circuit board 312a. A portion of the light-emitting element board 311 on the side of the flexible circuit board 312a and a portion of the flexible circuit board 312a on the side of the light-emitting element board 311 are bonded to the curved area 315c1 by the adhesive member E. This allows the curved area 315c1 to reliably fold the flexible circuit board 312a back into the space SP. The protective member 312d faces the light-emitting element board 311 and the portion of the flexible circuit board 312a corresponding to the curved area 315c1, and is curved along the curved area 315c1. As illustrated in FIGS. 4C and 4D, a fastening member SC1 (male screw) inserts through a through hole 312e1 of a retainer 312e and a through hole 312d1 of the protective member 312d on the one side L1 in the longitudinal direction L of the protective

member **312d** so that the protective member **312d** is fastened to an area to be fastened **315e** (female screw) of the main body member **315**. This allows the end of the protective member **312d** on the one side L1 in the longitudinal direction L to be fixed to the main body member **315**.

[0080] The fastening member SC1 inserts through the through hole **312d1** of the protective member **312d** on the other side L2 in the longitudinal direction L so that the protective member **312d** is fastened to the area to be fastened **315e** of the main body member **315**. This allows the end of the protective member **312d** on the other side L2 in the longitudinal direction L to be fixed to the main body member **315**. In the printed wiring board **312b**, the positioning hole **312b2** of the printed wiring board **312b** is positioned by the positioning protrusion **315b2** of the bent area **315b** on the other side L2 in the longitudinal direction L. In such a state, a fastening member SC2 (male screw) inserts through the through hole **315b3** of the bent area **315b** and a through recess **312b3** of the printed wiring board **312b** on the one side L1 in the longitudinal direction L so that the printed wiring board **312b** is fastened to an area to be fastened **312e2** (female screw) of the retainer **312e**. This allows the protective member **312d** to be held in position at the bent area **315b**.

[0081] The end of the flexible circuit board **312a** on the side of the printed wiring board **312b** disposed within the space SP is folded back at a corner between the placement area **315a** and the bent area **315b** and connected to the printed wiring board **312b**.

[0082] The drive circuit element **312c** is provided on the side of the placement area **315a** in the flexible circuit board **312a**. A recess **315a5** is provided on a facing surface **315a4** (inner surface) of the placement area **315a** facing the bent area **315b** to prevent interference with the drive circuit element **312c** in the flexible circuit board **312a**.

[0083] The main body member **315** includes a pair of side plates **315f** and **315g** (see FIGS. 4C and 4D) closing both ends of the space SP in the longitudinal direction L.

Tilt Adjustment Mechanism

[0084] Next, an example of the tilt adjustment mechanism **200** will be described below with reference to FIGS. 6A to 7.

[0085] FIG. 6A is a perspective view of a swing supporter **220** provided in the swinger **210**, as viewed obliquely from above on the back side X2. FIG. 6B is a perspective view illustrating a state in which the swing supporter **220** provided in the swinger **210** is inserted into a through hole FL2a in a main body frame FL2 on the back side X2, as viewed obliquely from above on the back side X2. FIG. 6C is an exploded perspective view of the swinger **210**, the swing supporter **220**, and the through hole FL2a in the main body frame FL2, as viewed obliquely from below on the back side X2. FIGS. 7A and 7B are perspective views of the tilt adjuster **230** provided on the main body frame FL1 on the front side X1, as viewed obliquely from above on the front side X1 and obliquely from below on the back side X2, respectively. FIGS. 7C and 7D are exploded perspective views of the tilt adjuster **230** as viewed obliquely from above on the front side X1 and the back side X2, respectively.

[0086] The tilt adjustment mechanism **200** includes the swinger **210**, the swing supporter **220** (see FIGS. 6A and 6B), and the tilt adjuster **230** (see FIGS. 7A to 7D). The print head **3** is provided on the swinger **210**. The swinger **210** is allowed to freely swing about the swing axis β by the swing supporter **220** provided on a swinger body **211**, and a tilt about the swing axis β is adjusted by the tilt adjuster **230**. The swing supporter **220** is fixed to an end of the swinger body **211** on the other side L2 (in such an example, the back side X2) in the longitudinal direction L. The main body frame FL (in such an example, the main body frame FL2 on the back side X2) (see FIG. 6B) of the image forming apparatus main body **101** (see FIG. 1) supports the swinger body **211** so that the swinger body **211** can freely swing around the swing axis β (see FIG. 6A).

[0087] In such an example, the main body frames FL1 and FL2 are supported at the centers in the rotation axis direction X by a main body frame FL3 aligned along the rotation axis direction X and the left-right direction Y. The swinger **210** is mounted on the main body frame FL3. The swing supporter **220** is a plate-shaped member extending in the optical axis direction N, as illustrated in

FIGS. 6A and 6B. The swing supporter **220** includes a protrusion **221** protruding toward the other side L2 and extending in the optical axis direction N. The main frame FL2 on the back side X2 includes a through hole FL2a through which the protrusion **221** passes.

[0088] As illustrated in FIG. 6C, the through hole FL2a supports swinging of the swing supporter **220** around the swing axis β by point contact at one or more (three in such an example) points. More specifically, one or more protrusions FL2a1 to FL2a3 are provided on the edge of the through hole FL2a of at least one side (both sides in such an example) in the width direction M. In such an example, two protrusions FL2a1 and FL2a2 are provided at an edge of the through hole FL2a on one side M1 in the width direction M, and two protrusions FL2a1 and FL2a2 are spaced apart in the up-down direction Z. One protrusion FL2a3 is provided on the edge of the through hole FL2a on the other side M2 in the width direction M between (specifically, at the center) the two protrusions FL2a1 and FL2a2 on the one side M1 provided in the up-down direction Z. A dimension g1 of the through hole FL2a in the width direction M (a dimension between the protrusions FL2a1 and FL2a2 on one side H1 and the protrusion FL2a3 on the other side in the width direction H) is larger than a dimension g2 of the protrusion **221** in the width direction M by a predetermined dimension to the extent that the swinger body **211** can swing. This allows the swinger body **211** to swing around the swing axis β .

[0089] The swing supporter **220** includes a plurality of (two) positioning holes **220a** and **220a** and a through hole **220b**. Positioning protrusions **211a** and **211a** (see FIG. 6C) and an area to be fastened **211b** (female screw) (see FIG. 6C) are provided on a rear surface **211e** of the swinger body **211** opposite the print head **3**. The positioning protrusions **211a** and **211a** are inserted into the plurality of positioning holes **220a** and **220a** of the swing supporter **220**. The area to be fastened **211b** is fastened by inserting a fastening member SC3 (male screw) (see FIG. 6C) through the through hole **220b** of the swing supporter **220**. This allows the swing supporter **220** to be fixed in position relative to the swinger body **211**.

[0090] As illustrated in FIGS. 7A to 7D, the tilt adjuster **230** includes a coupling member **231**, a moving member **232**, a biasing member **233**, and an adjustment member **234**. The coupling member **231** couples the swinger body **211** of the swinger **210** and the moving member **232**. The moving member **232** is provided movably on the one side M1 (left side Y1) and the other side M2 (right side Y2) in the width direction M relative to the main body frame FL1 on the front side of the image forming apparatus main body **101**. The biasing member **233** biases the moving member **232** toward the other side M2 in the width direction M (in such an example, the upstream side in the rotation direction R of the photosensitive drum **1** (see FIG. 1)) relative to the front main body frame FL1 on the front side. The adjustment member **234** presses the moving member **232** toward the one side M1 in the width direction M (in such an example, the downstream side in the rotation direction R of the photosensitive drum **1**) relative to the main body frame FL1 on the front side to adjust the position.

[0091] More specifically, the coupling member **231** is formed by stamping and bending a metal sheet, and includes a main body **2311**, and a first area to be fixed **231a** and a second area to be fixed **231b** coupled to the main body **2311**. The first area to be fixed **231a** is fixed to the swinger body **211** of the swinger **210**. The second area to be fixed **231b** is fixed to the moving member **232**.

[0092] The first area to be fixed **231a** includes a plurality of (two) positioning holes **231a1** and **231a1** and a through hole **231a2**. Protrusions **211c** and **211c** (see FIG. 7B) and the area to be fastened **211d** (female screw) (see FIG. 7B) are provided on the rear surface **211e** of the swinger body **211** opposite the print head **3**. The protrusions **211c** and **211c** are inserted into the plurality of positioning holes **231a1** and **231a1** of the first area to be fixed **231a**. The area to be fastened **211d** is fastened by inserting the fastening member SC4 (male screw) (see FIG. 7B) through the through hole **231a2** of the first area to be fixed **231a**. This allows the coupling member **231** to be fixed in position relative to the swinger **210**.

[0093] The second area to be fixed **231b** includes a plurality of (two) positioning holes **231b1** and

231b1, a through hole **231b2**, and a long through hole **231b3**. The moving member **232** includes positioning protrusions **232a** and **232a**, an area to be fastened **232b** (female screw) (see FIGS. 7B and 7D), and a long through hole **232c** (see FIGS. 7B and 7D). The positioning protrusions **232a** and **232a** are inserted into the plurality of positioning holes **231b1** and **231b1** of the second area to be fixed **231b**. The area to be fastened **232b** is fastened by inserting the fastening member SC5 (male screw) (see FIG. 7B) through the through hole **231b2** of the second area to be fixed **231b**. This allows the coupling member **231** to be fixed in position relative to the moving member **232**. The long through hole **231b3** and the long through hole **232c** are provided in the second area to be fixed **231b** and the moving member **232**, respectively, such that the long through hole **231b3** and the long through hole **232c** are aligned with each other.

[0094] A first guide **232d** (FIGS. 7B, 7C, and 7D) having a T-shape in cross section as viewed from the width direction M is provided on the side of the moving member **232** opposite to the positioning protrusions **232a** and **232a**. The main body frame FL1 on the front side includes a first slider FL1a (see FIGS. 7C and 7D). The first slider FL1a is a slit portion along the width direction M, and the first guide **232d** is inserted from the other side M2 to the one side M1 in the width direction M. This allows the moving member **232** to be freely movable in the width direction M. At an end of the moving member **232** on the one side M1 in the width direction M, a rod-shaped (cylindrical) second guide **232e** (see FIGS. 7A, 7C, and 7D) is provided along the width direction M. A first frame FL1b along the rotation axis direction X and the up-down direction Z of the main frame FL1 on the front side includes a second slider FL1b1. The second slider FL1b1 is a through hole along the width direction M, and the second guide **232e** is inserted therethrough. The biasing member **233** is a coil spring, and is inserted into the second guide **232e** between the first frame FL1b and the moving member **232**.

[0095] An area to be fastened FL1c1 (female screw) is provided on the second frame FL1c aligned along the rotation axis direction X and the up-down direction Z of the main frame FL1 on the front side. The adjustment member **234** is an adjustable fastener (a headless male screw, a so-called insect screw) along the width direction M. The area to be fastened FL1c1 is fastened to the adjustment member **234**. Thus, while the adjustment member **234** is biased toward the other side M2 in the width direction M by the biasing member **233**, an operator can rotate the adjustment member **234** with a tool such as a screwdriver to move the adjustment member **234** to the one side M1 or the other side M2 in the width direction M, to adjust a tilt of the print head **3** fixed to the moving member **232** about the swing axis β .

[0096] The main frame FL1 on the front side includes an area to be fastened FL1d (female screw) (see FIGS. 7C and 7D). The area to be fastened FL1d is fastened by inserting a fastening member SC6 (male screw) (see FIGS. 7C and 7D) through the long through hole **231b3** of the second area to be fixed **231b** and the long through hole **232c** of the moving member **232**. This allows the print head **3** to fix to the body frame FL1 on the front side after adjusting a tilt of the print head **3** around the swing axis β .

Positioning Mechanism

[0097] Next, an example of a positioning mechanism **300** will be described below with reference to FIGS. 8 to 12.

[0098] FIG. 8 is a perspective view of a photosensitive unit **400**, the print head **3**, and the swinger **210**, as viewed from above on the left side. FIG. 9A is a perspective view of a portion of a positioner **301a** on the one side L1 of the swinger **210** including the print head **3**, as viewed obliquely from above. FIGS. 9B and 9C are exploded perspective views of the print head **3** and the swinger **210** illustrated in FIG. 9A, as viewed obliquely from above and below, respectively. In FIGS. 9B and 9C, the coupling member **231**, the moving member **232**, and the adjustment member **234** are omitted in the figures. FIG. 9D is a cross-sectional perspective view of the positioner **301a** and the area to be positioned **302a** on the one side L1 illustrated in FIG. 8, as viewed obliquely from above. FIG. 9E is an enlarged view of the positioner **301a** and the area to be positioned **302a**

on the one side L1 in a $\gamma 1$ portion illustrated in FIG. 8.

[0099] FIG. 10A is a perspective view of a portion of a positioner **301b** on the other side L2 in the swinger **210** including the print head **3**, as viewed obliquely from above. FIGS. 10B and 10C are exploded perspective views of the print head **3** and the swinger **210** illustrated in FIG. 10A, as viewed obliquely from above and below, respectively. It is noted that the swing supporter **220** is omitted in FIGS. 10B and 10C. FIG. 10D is a cross-sectional perspective view of the positioner **301b** and the area to be positioned **302b** on the other side L2 illustrated in FIG. 8, as viewed obliquely from below. FIG. 10E is an enlarged view of the positioner **301b** and the area to be positioned **302b** on the other side L2 in a $\gamma 2$ portion illustrated in FIG. 8.

[0100] FIG. 11 is a cross-sectional view illustrating the positioner **301a** on the one side L1 abutting against the area to be positioned **302b** on the one side L1. FIG. 12 is a front view for explaining how the distance d between the print head **3** and the photosensitive drum **1** is maintained constant regardless of the tilt of the print head **3** around the swing axis β .

[0101] The image forming apparatus **100** further includes the positioning mechanism **300** (an example of a positioning mechanism). The positioning mechanism **300** positions the print head **3** relative to the photosensitive drum **1** to maintain a constant distance d (focal length) between the print head **3** and the photosensitive drum **1** regardless of the tilt of the print head **3** around the swing axis β caused by the tilt adjustment mechanism **200**. Here, “maintain a constant distance d ” has a concept including not only maintaining the focal length of the print head **3** constant, but also keeping the focal length of the print head **3** within a tolerance (depth of focus) (for example, ± 0.1 mm).

[0102] In the image forming apparatus **100** according to the present embodiment, as illustrated in FIG. 12, in adjusting the tilt of the print head **3** around the swing axis β , the print head **3** moves along a circumferential direction S of the photosensitive drum **1**, so that the distance d (d_1 , d_2 , d_3) between the photosensitive drum **1** and the print head **3** can be kept constant ($d_1=d_2=d_3$) depending on the position of the print head **3** in the circumferential direction S . This is more effective as the diameter (curvature) of the photosensitive drum **1** is smaller as the image forming apparatus **100** is more compact.

[0103] Thus, according to the present embodiment, regardless of the tilt of the print head **3** around the swing axis β caused by the tilt adjustment mechanism **200**, the positioning mechanism **300** can maintain a constant distance d between the print head **3** and the photosensitive drum **1**. This allows the distance d between the print head **3** and the photosensitive drum **1** to be constant, regardless of the tilt of the print head **3** around the swing axis β caused by the tilt adjustment mechanism **200**.

First Embodiment

[0104] In the present embodiment, the positioning mechanism **300** includes a positioner **301** (**301a**, **301b**) and an area to be positioned **302** (**302a**, **302b**). The positioner **301** (**301a**, **301b**) is provided in the print head **3** to determine a position of the print head **3** relative to the photosensitive drum **1**. The area to be positioned **302** (**302a**, **302b**) is provided on a main body side member (in such an example, a housing **401** of the photosensitive drum **1** (see FIG. 8)) which is a member on the main body side of the image forming apparatus **100** to abut against the positioner **301** (**301a**, **301b**). The area to be positioned **302** (in such an example, the area to be positioned **302a** on the one side L1) includes an arc-shaped area **3021** that is concentric with the center of rotation (rotation axis α) of the photosensitive drum **1** and has a radius larger than a radius r_1 of the photosensitive drum **1** (see FIG. 9D). More specifically, the area to be positioned **302** (**302a**) includes the arc-shaped area **3021** formed along a virtual arc δ having an arc radius r_2 (see FIG. 9D) larger than the radius r_1 of the photosensitive drum **1**, with the arc center being concentric with the center of rotation (α) of the photosensitive drum **1**. That is, the arc-shaped area **3021** has an arc radius r_2 (r_1+r_3) obtained by adding a predetermined distance r_3 (see FIG. 9D) to the radius r_1 of the photosensitive drum. The arc-shaped area **3021** has an abutment surface **3021a** that abuts against the positioner **301** (**301a**).

[0105] In such a configuration, even when the print head **3** swings around the swing axis β by the

tilt adjustment mechanism **200**, the positioner **301** (**301a**, **301b**) abuts against the abutment surface **3021a** (see FIGS. **9D** and **9E**) of the arc-shaped area **3021** of the area to be positioned **302**, so that the print head **3** can be positioned to maintain a constant distance *d* between the print head **3** and the photosensitive drum **1**. A configuration can be easily realized in which the distance *d* (see FIG. **12**) between the print head **3** and the photosensitive drum **1** is maintained constant by abutting the positioner **301** (**301a**, **301b**) against the area to be positioned **302** (**302a**, **302b**).

First Embodiment—1

[0106] In the present embodiment, the print head **3** includes the lens holding member **313** and the fixing member **314** (see FIGS. **5B**, **5C**, and the like). The lens holding member **313** holds the lens array **32**. The fixing member **314** fixes the light-emitting element board **311** or the panel. The positioner **301** (**301a**, **301b**) are provided in the lens holding member **313**.

[0107] In such a configuration, if the positioner **301** (**301a**, **301b**) is provided in the lens holding member **313** that holds lens array **32**, the position of print head **3** including the lens array **32** relative to the photosensitive drum **1** can be determined with high accuracy. Furthermore, there is no need to prepare a separate member for providing the positioner **301** (**301a**, **301b**), which makes it possible to realize a smaller and less costly image forming apparatus **100**.

First Embodiment—2

[0108] In the present embodiment, the positioning mechanism **300** further includes the biasing member **316** (**316a**, **316b**) that biases the positioner **301** (**301a**, **301b**) toward a main body side member (in such an example, the housing **401** of the photosensitive drum **1**). The area to be positioned **302** (**302a**, **302b**) is biased by the biasing member **316** (**316a**, **316b**) to abut against the positioner **301** (**301a**, **301b**). Here, the plurality of holders **315a3** and **315a3** in the lens holding member **313** including the positioner **301** (**301a**, **301b**) have an effect for restricting excessive movement of the fixing member **314** towards the main body side member (**401**) by the biasing member **316** (**316a**, **316b**).

[0109] In such a configuration, the biasing member **316** (**316a**, **316b**) enables the positioner **301** (**301a**, **301b**) reliably abutting against the area to be positioned **302** (**302a**, **302b**), and as a result, the distance *d* between the print head **3** and the photosensitive drum **1** can be stably maintained.

[0110] In such an example, the positioner **301** includes a pair of positioners **301a** and **301b**, one on each side of the print head **3** in the longitudinal direction *L*. The area to be positioned **302** includes a pair of areas to be positioned **302a** and **302b** corresponding to the pair of positioners **301a** and **301b**, respectively. The pair of positioners **301a** and **301b** are provided on both ends in the longitudinal direction *L* (outside the lens array **32**) in a surface **313b** of the lens holding member **313** on a side of the lens array **32**. Further, the biasing member **316** includes a pair of biasing members **316a** and **316b**.

[0111] The positioning mechanism **300** further includes a pair of lifting members **317a** and **317b**. The pair of lifting members **317a** and **317b** are provided at both ends of the swinger body **211** (outside the print head **3**) in the longitudinal direction *L* to be freely movable in the optical axis direction *N*. The print head **3** is provided to be freely movable in the optical axis direction *N* relative to the pair of lifting members **317a** and **317b**. The print head **3** is biased toward the photosensitive drum **1** by the pair of biasing members **316a** and **316b** arranged between the pair of lifting members **317a** and **317b** and the print head **3**, with movement to the one side *N1* in the optical axis direction *N* restricted by regulators **317a1** and **317b1**.

[0112] More specifically, an insertion protrusion (**317c**, **317d**), (**317c**, **317d**) is provided on a surface **317g** of the pair of lifting members **317a** and **317b** on a side of the swinger **210**. The insertion protrusion (**317c**, **317d**), (**317c**, **317d**) extends toward a side of the print head **3** (the one side *N1*) in the optical axis direction *N*. The rear surface **314b** of the fixing member **314** includes an insertion recess (**314d**, **314e**), (**314d**, **314e**) (see FIGS. **9C** and **10C**) into which the insertion protrusion (**317c**, **317d**), (**317c**, **317d**) is inserted, respectively. This allows the print head **3** to be restricted in the longitudinal direction *L* and width direction *M* relative to the pair of lifting

members **317a** and **317b** while allowing the print head **3** to be freely moved in the optical axis direction N relative to the pair of lifting members **317a** and **317b**. The pair of biasing members **316a** and **316b** are coil springs into which the insertion protrusions **317c** and **317c** on the one side M1 is inserted between the fixing member **314** and the pair of lifting members **317a** and **317b**. This allows the print head **3** to be biased toward the photosensitive drum **1** for the pair of lifting members **317a** and **317b**.

[0113] The pair of lifting members **317a** and **317b** include the regulators **317a1** and **317b1**. The pair of lifting members **317a** and **317b** and the regulators **317a1** and **317b1** are integrally formed. The regulators **317a1** and **317b1** detachably restrict movement of the fixing member **314** of the print head **3** to the one side N1 in the optical axis direction N. The regulators **317a1** and **317b1** include lockers **317a11** and **317b11** that lock the surface **314a** of the fixing member **314**.

[0114] The photosensitive drum **1** and the print head **3** are close to each other, and thus, when an operator replaces the photosensitive drum **1**, the photosensitive drum **1** may come into contact with the print head **3** in inserting or removing the photosensitive drum **1** in the rotation axis direction X relative to the image forming apparatus main body **101**.

[0115] In this regard, the positioning mechanism **300** further includes an actuating member **318** that raises and lowers the pair of lifting members **317a** and **317b**. The actuating member **318** is movable in the longitudinal direction L relative to the swinger body **211**, and movement in the longitudinal direction L raises and lowers the pair of lifting members **317a** and **317b**.

[0116] More specifically, the surface **211f** of the swinger body **211** on the side of the print head **3** includes an insertion protrusion (**211g**, **211g**), (**211g**, **211g**) erected at both ends in the longitudinal direction L. The insertion protrusion (**211g**, **211g**), (**211g**, **211g**) extends toward the one side N1 in the optical axis direction N. The rear surface **317e** of the pair of lifting members **317a** and **317b** on a side of the swinger **210** includes an insertion recess (**317f**, **317f**), (**317f**, **317f**) (see FIGS. 9C and 10C) into which the insertion protrusion (**211g**, **211g**), (**211g**, **211g**) is respectively inserted. This allows the pair of lifting members **317a** and **317b** to be restricted in the longitudinal direction L and width direction M relative to the swinger body **211** while allowing the pair of lifting members **317a** and **317b** to be freely movable in the optical axis direction N relative to the swinger body **211**.

[0117] The actuating member **318** is provided between the pair of lifting members **317a** and **317b** and the surface **211f** of the swinger body **211**. The actuating member **318** includes a plurality of (two) tilted surfaces **318a** and **318a** and top surfaces **318b** and **318b**. The tilted surfaces **318a** and **318a** are tilted to slide against the pair of lifting members **317a** and **317b** as the actuating member **318** moves to either one side L1 or the other side L2 in the longitudinal direction L, and as a result, the pair of lifting members **317a** and **317b** is tilted to be moved away from and towards the photosensitive drum **1**, respectively. The pair of lifting members **317a** and **317b** include tilted surfaces **317h** and **317h** (see FIGS. 9B and 10B). The tilted surfaces **317h** and **317h** are provided at locations corresponding to the tilted surfaces **318a** and **318a** in the actuating member **318** of the pair of lifting members **317a** and **317b**, and are tilted along the tilted surfaces **318a** and **318a**.

[0118] In the configuration described above, when an operator moves the actuating member **318** from a first position where the pair of lifting members **317a** and **317b** are farthest from the photosensitive drum **1** to the other side L2 in the longitudinal direction L, the tilted surfaces **318a** and **318a** slide against the tilted surfaces **317h** and **317h** of the pair of lifting members **317a** and **317b**. At this time, the biasing forces of the biasing members **316a** and **316b** act on the pair of lifting members **317a** and **317b**, causing the pair of lifting members **317a** and **317b** to approach the photosensitive drum **1** (rise). When the actuating member **318** moves further toward the other side L2 in the longitudinal direction L and the top surfaces **318b** and **318b** contact the rear surfaces **317e** of the pair of lifting members **317a** and **317b**, the pair of lifting members **317a** and **317b** are maintained in a second position closest to the photosensitive drum **1**. This brings the print head **3** close to the photosensitive drum **1** at a predetermined distance d to make the print head **3** ready for image formation.

[0119] On the other hand, when the operator moves the actuating member **318** from the second position of the pair of lifting members **317a** and **317b** to the one side L1 in the longitudinal direction L, the top surfaces **318b** and **318b** of the actuating member **318** move away from the rear surfaces **317e** of the pair of lifting members **317a** and **317b**, and the tilted surfaces **318a** and **318a** slide against the tilted surfaces **317h** and **317h** of the pair of lifting members **317a** and **317b**. At this time, the biasing forces applied by the biasing members **316a** and **316b** to the pair of lifting members **317a** and **317b** are released, and the pair of lifting members **317a** and **317b** move away from the photosensitive drum **1** (descend). When the actuating member **318** moves further toward the one side L1 in the longitudinal direction L, the pair of lifting members **317a** and **317b** return to the first position. This makes it possible to prevent the photosensitive drum **1** from coming into contact with the print head **3** when the operator replaces the photosensitive drum **1**.

First Embodiment—3

[0120] In the present embodiment, the positioner **301a** on the one side L1 (the front side X1 in such an example) of the pair of positioners **301a** and **301b** includes a convex area **301a1** extending in the longitudinal direction L. Of the pair of areas to be positioned **302a** and **302b**, the area to be positioned **302a** on the one side L1 corresponding to the positioner **301a** on the one side L1 includes an arc-shaped area **3021**. The arc-shaped area **3021** includes an abutment surface **3021a** abutting against the convex area **301a1** of the positioner **301a** on the one side L1. In such an example, the convex area **301a1** has a mountain-like shape as viewed in cross section from the rotation axis direction X.

[0121] In such a configuration, the convex area **301a1** abutting against the arc-shaped area **3021** extends in the longitudinal direction L, so that the convex area **301a1** can be brought into stable and reliable abutment against the arc-shaped area **3021**.

First Embodiment—4

[0122] In the present embodiment, the positioner **301b** on the other side L2 (in such an example, the back side X2) of the pair of positioners **301a** and **301b** includes a swing shaft **301b1** rotating about the swing axis β (see FIGS. **3** and **6A**). The print head **3** is pivotable about the swing shaft **301b1**. Of the pair of areas to be positioned **302a** and **302b**, the area to be positioned **302b** on the other side L2 corresponding to the positioner **301b** on the other side L2 includes a concave area **302b1** (groove) extending in the longitudinal direction L (see FIGS. **10D** and **10E**). The concave area **302b1** fittingly receives the swing shaft **301b1** in the positioner **301b** on the other side L2.

[0123] In such a configuration, the concave area **302b1** in the area to be positioned **302b** on the other side L2 fittingly receives the swing shaft **301b1** in the positioner **301b** on the other side L2 to allow some leeway in a mounting position of the print head **3** to the main body side member (**401**) in the longitudinal direction L. Even if the print head **3** deviates slightly in the longitudinal direction L from a reference position relative to the main body side member (**401**) within an allowable range, the deviation of the image in the longitudinal direction L can ultimately be corrected in image processing.

[0124] The positioner **301** is the pair of positioners **301a** and **301b**, one on each side of the print head **3** in the longitudinal direction L. The area to be positioned **302** includes a pair of areas to be positioned **302a** and **302b** corresponding to the pair of positioners **301a** and **301b**, respectively. This allows the pair of positioners **301a** and **301b** to more reliably abut against the areas to be positioned **302a** and **302b** by the biasing members **316a** and **316b** to make it possible to keep the distance *d* between the print head **3** and the photosensitive drum **1** constant in an even more stable manner.

First Embodiment—5

[0125] FIG. **13A** is an exploded perspective view of the positioner **301a** on the one side L1 and the lens holding member **313** as viewed obliquely from below. FIG. **13B** is a perspective view illustrating the positioner **301a** on the one side L1 attached to the lens holding member **313** as viewed obliquely from below. FIG. **14A** is a perspective view of the positioner **301a** on the one

side L as viewed from above on the back side X2. FIG. 14B is a perspective view of the positioner **301a** on the one side L1 as viewed from below on the front side X1.

[0126] FIG. 15A is an exploded perspective view of the positioner **301b** on the other side L2 and the lens holding member **313**, as viewed obliquely from below. FIG. 15B is a perspective view illustrating the positioner **301b** on the other side L2 attached to the lens holding member **313** as viewed obliquely from below. FIG. 16A is a perspective view of the positioner **301b** on the other side L2 as viewed from above on the back side X2. FIG. 16B is a perspective view of the positioner **301b** on the other side L2 as viewed from below on the front side X1.

[0127] In the present embodiment, the positioner **301** (**301a**, **301b**) is detachably provided on the lens holding member **313** of the print head **3**. The positioner **301** (**301a**, **301b**) includes a slip-out preventor **3011** that prevents the positioner **301** from slipping out of the lens holding member **313**.

[0128] In such a configuration, the positioner **301** (**301a**, **301b**) is detachably provided on the lens holding member **313**, and as a result, the ease of attaching and detaching the positioner **301** (**301a**, **301b**) to and from the lens holding member **313** can be improved. If the positioner **301** (**301a**, **301b**) includes the slip-out preventor **3011**, it is possible to effectively prevent the positioner **301** (**301a**, **301b**) from easily coming off the lens holding member **313**.

[0129] In such an example, the positioner **301** (**301a**, **301b**) is detachable by moving in the longitudinal direction L relative to the lens holding member **313**. The lens holding member **313** includes a through hole penetrating in the optical axis direction N (thickness direction) or a recess recessed in the optical axis direction N (through hole **3131** in such an example). The slip-out preventor **3011** in the positioner **301** (**301a**, **301b**) includes a protrusion **3011a** protruding toward a side of the lens holding member **313** in the optical axis direction N (the other side N2). The protrusion **3011a** is fittingly received into a through hole or a recess (through hole **3131**) in the lens holding member **313**.

[0130] In such a configuration, the positioner **301** (**301a**, **301b**) is detachable by moving in the longitudinal direction L relative to the lens holding member **313**, to make it easy to attach and detach the positioner **301** (**301a**, **301b**) in the longitudinal direction L relative to the lens holding member **313**. If the protrusion **3011a** is provided on the positioner **301** (**301a**, **301b**) and a through hole or a recess (through hole **3131**) is provided in the lens holding member **313**, a configuration in which the positioner **301** (**301a**, **301b**) is prevented from coming off the lens holding member **313** can be easily realized.

[0131] More specifically, the positioner **301** (**301a**, **301b**) includes a rod-shaped section **3012** extending in the optical axis direction N, and a pair of flat plates **3013** and **3014** provided on both ends of the rod-shaped section **3012** in the optical axis direction N. In the positioner **301a** on the one side L1 in the longitudinal direction L, a convex area **301a1** is provided in the flat plate **3013** on the one side N1 in the optical axis direction N, opposite to the rod-shaped section **3012**. In the positioner **301b** on the other side L2 in the longitudinal direction L, the swing shaft **301b1** is provided in the flat plate **3013** on the one side N1 in the optical axis direction N, opposite to the rod-shaped section **3012**. Sizes h1 and h2 (see FIGS. 13A and 15A) of the flat plates **3013** and **3014** in the width direction M in the optical axis direction N are larger than a size h3 (see FIGS. 13A and 15A) of the rod-shaped section **3012** in the width direction M ($h1 > h3$, $h2 > h3$).

[0132] The area to be positioned **302** (**302a**, **302b**) includes a first insertion area **3132** through which the rod-shaped section **3012** is inserted in the longitudinal direction L, and a second insertion area **3133** through which the flat plate **3014** on the other side N2 is inserted in the longitudinal direction L. A size h4 of the second insertion area **3133** in the width direction M (see FIGS. 13A and 15A) is larger than the size h2 of the flat plate **3014** in the width direction M on the other side N2 by a predetermined size ($h4 > h2$) so that the flat plate **3014** on the other side N2 can be smoothly inserted therethrough. A size h5 of the first insertion area **3132** in the width direction M (see FIGS. 13A and 15A) is larger than the size h3 of the rod-shaped section **3012** in the width direction M by a predetermined size ($h5 > h3$) so that the rod-shaped section **3012** can be smoothly

inserted therethrough. The size h_4 in the width direction M of the second insertion area **3133** is larger than the size h_5 in the width direction M of the first insertion area **3132** ($h_4 > h_5$).

[0133] Thus, if the positioner **301** (**301a**, **301b**) is moved in the longitudinal direction L relative to the lens holding member **313** and the protrusion **3011a** is fitted into the through-holes **3131**, it is possible to effectively prevent the positioner **301** (**301a**, **301b**) from coming off the lens holding member **313**.

Second Embodiment

[0134] FIG. **17** is a side view illustrating another example of the positioning mechanism **300**. FIGS. **18A** and **18B** are front and rear views of a guide mechanism **330** illustrated in FIG. **17**, respectively. The tilt adjuster **230** is omitted from illustration in FIGS. **17** to **18B**. FIG. **19** is a front view for explaining that regardless of the tilt of the print head **3** around the swing axis β , a distance d between the print head **3** and the photosensitive drum **1** is maintained constant, and the optical axes of the plurality of light-emitting elements **31** to **31** in the print head **3** orient toward the center of rotation (α) of the photosensitive drum **1**.

[0135] The second embodiment is similar to the first embodiment except that the swing supporter **220** is removed and the guide mechanism **330** is provided in the first embodiment, and the same components are given the same reference numerals and the description of the same components will be omitted.

[0136] In the image forming apparatus **100** according to the first embodiment, as the print head **3** tilts around the swing axis β , the optical axes of the plurality of light-emitting elements **31** to **31** in the print head **3** move in parallel. The surface **1a** of the photosensitive drum **1** is curved, and thus, depending on the position of light emitted by the plurality of light-emitting elements **31** to **31** on the surface **1a** of the photosensitive drum **1**, a beam spot on the surface **1a** of the photosensitive drum **1** may not have the desired shape (preferably, be circular), which results in a deterioration in image quality.

[0137] In this regard, in the present embodiment, the positioning mechanism **300** positions the print head **3** relative to the photosensitive drum **1** so that the optical axes of the plurality of light-emitting elements **31** to **31** in the print head **3** orient toward the center of rotation (α) of the photosensitive drum **1** regardless of the tilt of the print head **3** around the swing axis β caused by the tilt adjustment mechanism **200**. Here, “the optical axes of the plurality of light-emitting elements **31** to **31** orient toward the center of rotation (α) of the photosensitive drum **1**” is a concept including not only a case where the optical axes of the light-emitting elements **31** to **31** pass through the center of rotation (α) of the photosensitive drum **1**, but also a case where a beam spot irradiated onto the surface **1a** of the photosensitive drum **1** has an acceptable shape (a shape in which the image quality is maintained).

[0138] In such a configuration, regardless of the tilt of the print head **3** around the swing axis β , the optical axes of the plurality of light-emitting elements **31** to **31** are oriented toward the center of rotation (α) of the photosensitive drum **1**. Therefore, regardless of the position of light emitted by the plurality of light-emitting elements **31** to **31** on the surface **1a** of the photosensitive drum **1**, a beam spot on the surface **1a** of the photosensitive drum **1** can be made into a desired shape (preferably, be circular), as a result of which it is possible to effectively prevent a degradation in image quality.

Second Embodiment—1

[0139] In the present embodiment, the positioning mechanism **300** includes the guide mechanism **330** (an example of a guiding mechanism). When the print head **3** is tilted around the swing axis β by the tilt adjustment mechanism **200**, while the guide mechanism **330** maintains the distance d between the print head **3** and the photosensitive drum **1**, the guide mechanism **33** guides the print head **3** along the virtual arc δ so that a virtual straight line ϕ (see FIG. **18A**) along the board surface **311a** of the light-emitting element board **311** is a tangent to the virtual arc δ with an arc radius r_2 ($r_1 + r_3$) larger than the radius r_1 of the photosensitive drum **1** and with the arc center concentric

with the center of rotation (α) of the photosensitive drum **1**.

[0140] In such a configuration, the guide mechanism **330** that guides the print head **3** along the virtual arc δ can be used to easily realize a configuration in which the optical axes of the plurality of light-emitting elements **31** to **31** in the print head **3** are oriented toward the center of rotation (α) of the photosensitive drum **1**.

Second Embodiment—2

[0141] In the present embodiment, the guide mechanism **330** includes a print head guide **331** and a print head pivotter **332**. The print head guide **331** guides the one side L1 in the longitudinal direction L of the print head **3** relative to the image forming apparatus main body **101** along the virtual arc δ . The print head pivotter **332** allows the other side L2 in the longitudinal direction L of the print head **3** to pivot (be freely rotatable) about a pivot axis λ (see FIG. **18B**) along the longitudinal direction L relative to the image forming apparatus main body **101**.

[0142] In such a configuration, the print head **3** capable of pivoting around the pivot axis λ on the other side L2 in the longitudinal direction L by the print head pivotter **332**, can be guided along the virtual arc δ on the one side L1 in the longitudinal direction L by the print head guide **331**. This allows the print head **3** to be moved stably along the virtual arc δ .

[0143] More specifically, the print head guide **331** includes a guiding member **3311** and a guided member **3312**. The guiding member **3311** is fixed to an end of the print head **3** on the one side L1 in the longitudinal direction L (in such an example, the rear surface **211e** of the swinger body **211**) by a fastening member (male screw) (not illustrated) or the like.

[0144] The guiding member **3311** is formed by stamping and bending a metal sheet, and includes a guiding member main body **3311a** and a bent area **3311b**. The guiding member body **3311a** is fixed to the print head **3** and extends from the print head **3** toward the one side L1 in the longitudinal direction L. The bent area **3311b** is bent in the up-down direction Z (downward in such an example) from the tip of the guiding member main body **3311a**. The bent area **3311b** includes an arc-shaped through-hole **3311c** (see FIG. **18A**) formed along the virtual arc δ . The image forming apparatus main body **101** (in such an example, the main body frame FL1 on the front side X1) includes the guided member **3312** to be inserted into the arc-shaped through hole **3311c**.

[0145] The guided member **3312** is a cylindrical member provided in the image forming apparatus main body **101**, and in such an example, the guided member **3312** is fixed (in such an example, fixed by crimping using a crimping process utilizing the plastic deformation of metal) to the main body frame FL1 on the front side X1 so that the axis direction of the guided member **3312** is along the rotation axis direction X of the photosensitive drum **1**. The guided member **3312** protrudes from the main frame FL1 toward the front side X1. In the bent area **3311b** of the guiding member **3311**, the guided member **3312** is inserted into the arc-shaped through hole **3311c** at a position on the front side X1 from the main frame FL1. Although the guided member **3312** has a circular shape in cross section as viewed from the axis direction, the guided member **3312** may be formed into a shape conforming to the arc-shaped through hole **3311c**.

[0146] The print head pivotter **332** includes a support member **3321** and a shaft member **3322**. The support member **3321** is fixed to the end of the print head **3** on the other side L2 in the longitudinal direction L (in such an example, the rear surface **211e** of the swinger body **211**) by a fastening member (male screw) (not illustrated) or the like.

[0147] The support member **3321** is formed by stamping and bending a metal sheet, and includes a support member main body **3321a** and a bent area **3321b**. The support member main body **3321a** is fixed to the print head **3** and extends from the print head **3** toward the one side L2 in the longitudinal direction L. The bent area **3321b** is bent in the up-down direction Z (downward in such an example) from the tip of the support member main body **3321a**. The image forming apparatus main body **101** (in such an example, the main body frame FL2 on the back side X2) includes a circular through-hole FL2b (see FIG. **17**) penetrating in a circular shape. The bent area **3321b** includes a shaft member **3322** to be inserted into the circular through hole FL2b.

[0148] The shaft member **3322** is a cylindrical member provided at the bent area **3321b**, and in such an example, the shaft member **3322** is fixed (in such an example, fixed by crimping) to the main frame **FL2** so that the axis direction of the shaft member **3322** is along the rotation axis direction **X** of the photosensitive drum **1**. The bent area **3321b** of the support member **3321** is located on the front side **X1** (inner side) from the main frame **FL2**. The shaft member **3322** is inserted into the circular through-hole **FL2b** in the main frame **FL2** and protrudes from the main frame **FL2** toward the back side **X2**.

[0149] In the configuration described above, the guide mechanism **330** can orient the optical axes of the plurality of light-emitting elements **31** to **31** in the print head **3** toward the center of rotation (α) of the photosensitive drum **1**, regardless of the tilt of the print head **3** around the swing axis β .
Third Embodiment

[0150] In the present embodiment, the main body side member in the first and second embodiments is a housing (in such an example, a housing **401** of the photosensitive drum **1**) that rotatably supports the photosensitive drum **1** around the rotation axis α of the photosensitive drum **1** (see FIG. **8**). The area to be positioned **302** (**302a**, **302b**) is provided integrally with the housing (**401**). That is, the area to be positioned **302** (**302a**, **302b**) is integrally provided with the housing (**401**).

[0151] In such a configuration, the area to be positioned **302** (**302a**, **302b**) is provided integrally with the housing (**401**) of the photosensitive drum **1**, so that there is no need to prepare a separate member for providing the area to be positioned **302** (**302a**, **302b**) to realize a more compact and less costly image forming apparatus **100**.

[0152] More specifically, the area to be positioned **302** (**302a**, **302b**) is integrally formed with the housing **401** (see FIG. **8**) on both outer sides of the photosensitive drum **1** in the rotation axis direction **X** in the photosensitive unit **400**. The housing of the photosensitive drum **1** may be provided in the image forming apparatus main body **101**.

[0153] The present disclosure is not limited to the embodiments described above and can be embodied in various other forms. Therefore, the present embodiment is to be construed in all respects as merely illustrative and not restrictive. The scope of the present disclosure is defined by the claims and is not limited in any way by the text of the specification. Furthermore, all modifications and variations within a scope equivalent to the scope of the claims are within the scope of the present disclosure.

Claims

1. An image forming apparatus comprising: an image carrier; a print head having a long length and including a board including a plurality of light-emitting elements arranged in a line and a lens array that focuses light emitted by the plurality of light-emitting elements onto the image carrier; and a tilt adjuster that adjusts a tilt of the print head about a swing axis along an optical axis direction of the plurality of light-emitting elements, wherein the image forming apparatus further includes a positioning mechanism that positions the print head relative to the image carrier to maintain a constant distance between the print head and the image carrier, regardless of the tilt of the print head about the swing axis caused by the tilt adjuster.
2. The image forming apparatus according to claim 1, wherein the positioning mechanism includes: a positioner that is provided in the print head and determines a position of the print head relative to the image carrier; and an area to be positioned that is provided in a main body side member being a member on a side of an image forming apparatus main body and abuts against the positioner, wherein the area to be positioned includes an arc-shaped area concentric with a center of rotation of the image carrier and larger than a radius of the image carrier, and the arc-shaped area includes an abutment surface abutting against the positioner.
3. The image forming apparatus according to claim 2, wherein the print head includes a lens

holding member that holds the lens array and a fixing member that fixes the board or a panel, and the positioner is provided in the lens holding member.

4. The image forming apparatus according to claim 2, wherein the positioning mechanism further includes a biasing member that biases the positioner toward the main body side member, and the area to be positioned is biased by the biasing member to abut against the positioner.

5. The image forming apparatus according to claim 4, wherein the positioner includes a pair of positioners, one on each side of the print head in a longitudinal direction, the area to be positioned includes a pair of areas to be positioned corresponding to the pair of positioners, one of the pair of positioners includes a convex area extending in the longitudinal direction, and the area to be positioned on one side corresponding to the positioner on the one side, of the pair of areas to be positioned, includes the arc-shaped area, and the abutment surface of the arc-shaped area abuts against the convex area of the positioner on the one side.

6. The image forming apparatus according to claim 4, wherein the positioner includes a pair of positioners, one on each side of the print head in a longitudinal direction, the area to be positioned includes a pair of areas to be positioned corresponding to the pair of positioners, the positioner on the other side, of the pair of positioners, includes a swing shaft centered on the swing axis, and the print head is pivotable about the swing shaft, and the area to be positioned on the other side corresponding to the positioner on the other side, of the pair of areas to be positioned, includes a concave area extending in the longitudinal direction, and the concave area fittingly receives the swing shaft of the positioner on the other side.

7. The image forming apparatus according to claim 3, wherein the positioner is detachably provided in the lens holding member in the print head, and the positioner includes a slip-out preventor that prevents the positioner from coming off the lens holding member.

8. The image forming apparatus according to claim 7, wherein the positioner is detachable from the lens holding member in a longitudinal direction of the print head, the lens holding member includes a through hole penetrating in the optical axis direction or a recess recessed in the optical axis direction, and the slip-out preventor in the positioner includes a protrusion protruding toward the lens holding member in the optical axis direction, and the protrusion is fittingly received into the through hole or the recess in the lens holding member.

9. The image forming apparatus according to claim 2, wherein the main body side member is a housing that rotatably supports the image carrier around a rotation axis of the image carrier, and the area to be positioned is integrated with the housing.

10. The image forming apparatus according to claim 1, wherein the positioning mechanism positions the print head relative to the image carrier so that optical axes of the plurality of light-emitting elements in the print head are oriented toward a center of rotation of the image carrier, regardless of the tilt of the print head about the swing axis caused by the tilt adjuster.

11. The image forming apparatus according to claim 10, wherein the positioner includes a guiding mechanism that guides the print head along a virtual arc when the print head is tilted about the swing axis by the tilt adjuster while maintaining a distance between the print head and the image carrier, so that a virtual straight line along a board surface of the board is a tangent to the virtual arc, a center of the virtual arc is concentric with a center of rotation of the image carrier, and an arc radius of the virtual arc is larger than a radius of the image carrier.

12. The image forming apparatus according to claim 11, wherein the guiding mechanism includes a print head guide that guides one side of the print head in a longitudinal direction relative to an image forming apparatus body along the virtual arc, and a print head pivotter that allows the other side of the print head in the longitudinal direction to be pivotable about a pivot axis along the longitudinal direction relative to the image forming apparatus body.

13. The image forming apparatus according to claim 1, wherein the board is a film-like board including an organic EL diode element.
